Nina Mason

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SUMMARY

Software engineer with hands-on experience building AI-powered, user-focused applications using Python, Java, React, MongoDB, and Streamlit. Experienced in Agile software development, testing, and shipping full-stack features in collaborative remote environments.

EDUCATION

B.S.E., Software Engineering, Summa Cum Laude May 2026
Arizona State University, Tempe, AZ 3.89 GPA
Relevant Coursework: Data Structures & Algorithms, Secure Software Systems, Web-Based Applications, Operating Systems

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, C, C++, C#, HTML/CSS
Frameworks & Libraries: React, Node.js, JavaFX, Swing, Tkinter, Streamlit
Databases: MongoDB Atlas, MySQL, SQL (query writing, joins), JSON-based data persistence
Tools & Build Systems: Git/Github, Gradle, Docker, YAML prompt configuration, CI/CD, Figma, Astah, Lucidchart, Taiga
Back-End & APIs: REST APIs, LLM prompt engineering, multi-step reasoning pipelines
Operating Systems: Windows, macOS, Linux
Additional: Troubleshooting, debugging, version control, object-oriented backend design, Arduino (sensor integration), MIPS Assembly

PROFESSIONAL EXPERIENCE

Self-Learning AI Tutor—Capstone Project (Sponsored by MyEdMaster LLC) Aug 2025 – May 2026

- Designed and implemented an adaptive learning algorithm using Python, Streamlit, and MongoDB to dynamically select questions based on student performance
- Developed and maintained unit and integration tests (pytest), improving test coverage and overall system reliability
- Built and enhanced production-facing features for the AI-powered tutoring platform, including a crash course learning module and structured reasoning analysis workflows
- Collaborated in a fully remote Agile Scrum team, participating in sprint planning, stand-ups, and code reviews to deliver iterative features

PROJECTS

GPS Distance App, *Personal Project* Sep 2025 – Present

- Built CLI and JavaFX GUI applications for geospatial distance and travel-time calculations using the haversine formula
- Integrated a Mapbox API and embedded dynamic HTML maps into JavaFX, debugging API and UI rendering issues to ensure accurate visualization
- Designed intuitive route creation and selection workflows and automated builds using Gradle
- Continuously enhance the application with new features, including route visualization improvements and image export functionality

Memoranda Software to Bus Scheduler Transformation, *Class Project* Jan 2025 – Mar 2025

- Extended a legacy Java codebase within a Scrum team to support new bus routing functionality, adapting existing architecture to meet evolving client needs
- Implemented backend routing logic and contributed to a GPS-based mapping UI, refactoring fragile components to improve system stability and usability

Image Filter Processor, *Class Project* Oct 2024 – Dec 2024

- Implemented a pixel-by-pixel BMP image processing pipeline in C, requiring careful memory management and low-level data handling
- Parallelized image filtering using pthreads, optimizing thread coordination and reconstruction logic to improve performance without corrupting image output

WORK EXPERIENCE

Starbucks, Loveland, CO: Barista May 2018 – April 2026

- Demonstrated leadership and operational reliability in a high-volume environment, mentoring 10+ new hires, resolving time-sensitive issues, leading peak shifts, maintaining performance under pressure, and earning *Partner of the Quarter* recognition